

Multiple Choice (10 marks)

1. Which of the following adaptive features would least likely be found in an animal living in a hot arid environment?
 - (a) High metabolic rate to generate heat
 - (b) Storage of water in fatty tissues
 - (c) Long loops of Henle to maximize water reabsorption
 - (d) Short loops of Henle to maximize water secretion
 - (e) Large amounts of body hair to trap heat
2. During the courtship of the common tern, the male presents a fish to another tern. Account for this behavior and discuss why courtship in general may be necessary for reproduction.
 - (a) Courtship is primarily for entertainment purposes among terns
 - (b) Courtship is not necessary for reproduction
 - (c) Courtship is a way to show dominance
 - (d) Courtship is a random behavior with no impact on reproduction
 - (e) Courtship is necessary for physical transformation required for reproduction
3. What function does oxygen serve in the body?
 - (a) Oxygen is used to maintain the temperature of cells.
 - (b) Oxygen is used by cells to produce alcohol.
 - (c) Oxygen is used to produce carbon dioxide in cells.
 - (d) Oxygen is solely responsible for stimulating brain activity.
 - (e) Oxygen is used to synthesize proteins in cells.
4. Which of the following is the source of oxygen produced during photosynthesis?
 - (a) O_2
 - (b) CH_4 (Methane)
 - (c) CO_2
 - (d) H_2O
 - (e) N_2 (Nitrogen)
5. What is the genetic basis of Down's syndrome?
 - (a) Duplication of chromosome 14

- (b) Translocation between chromosomes 14 and 21
- (c) Deletion in chromosome 21
- (d) Mutation in chromosome 14
- (e) Duplication of chromosome 18

True / False (10 marks)

Answer True or False

- 6. True or False: Glycolysis produces NADP⁺, ADP, and sugar as its main end products during cellular respiration.
- 7. True or False: Comparing gene frequencies and genotype frequencies over two generations can determine if a population is in genetic equilibrium.
- 8. True or False: The probability of getting exactly three heads in five flips of a balanced coin is 25 percent.
- 9. True or False: A photosynthetic plant growing in iron- and magnesium-deficient soil would appear pale, small, short, and weak due to nutrient deficiency.
- 10. True or False: Cholesterol uptake relies heavily on adhesion plaques for effective cellular internalization.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the genetic inheritance of hemophilia in a couple where one son is affected by this recessive sex-linked disorder. Analyze the probability of their next daughter also having hemophilia.
- 12. Discuss the concept of coevolution, particularly in the context of predator-prey relationships, and analyze its implications for biodiversity and ecosystem dynamics.

End of Paper